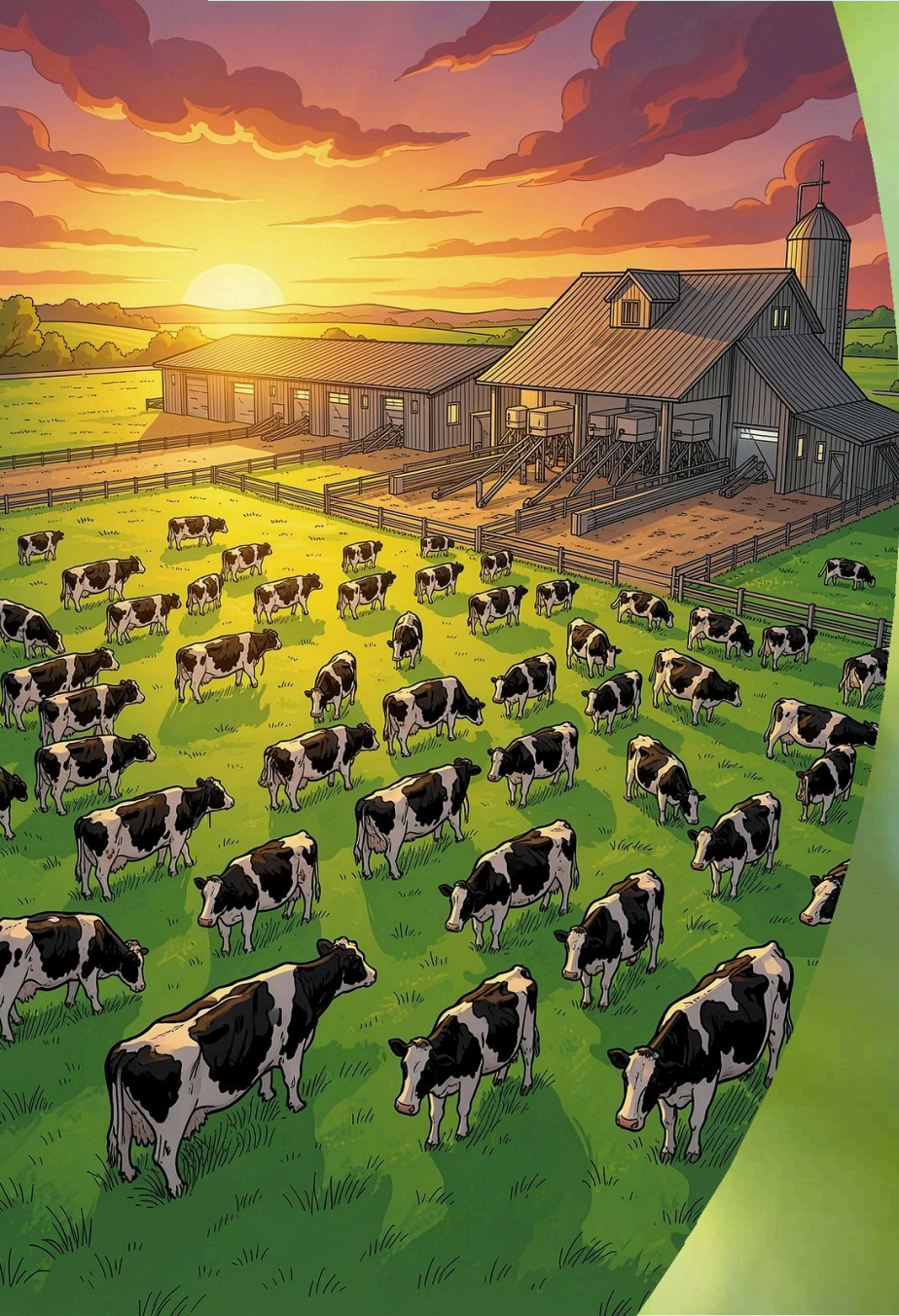




DBMS PROJECT · REVIEW 1

Dairy Farm Herd & Milk Collection Management System

Database Management System —Review 1 Presentation

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Dairy Operations Generate Data Everywhere.

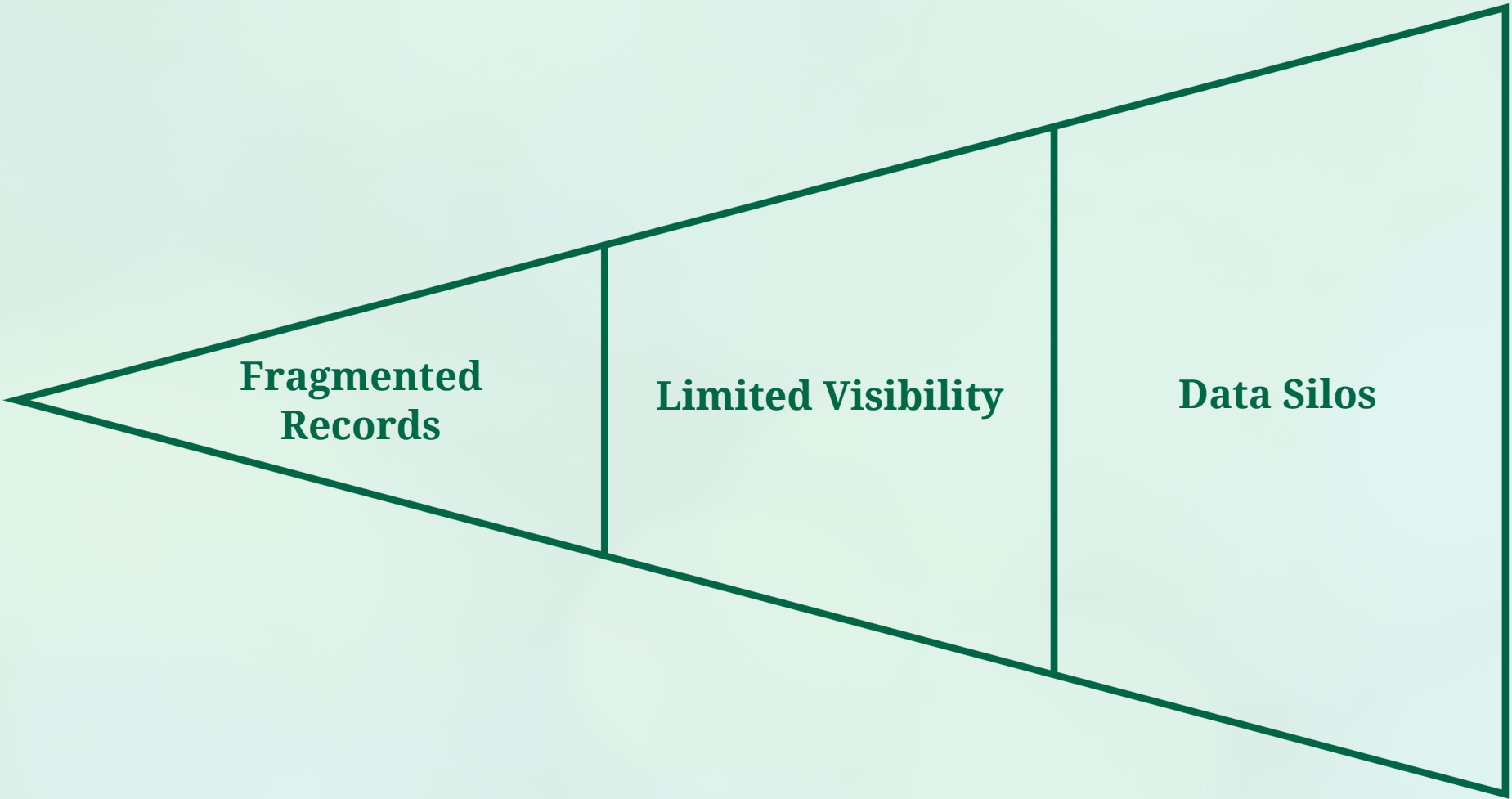
Fragmented manual records create data silos across animal health, feed, milk, and sales—duplication, inconsistency, and limited traceability follow.

Animal Records
Breed, health, and status maintained separately

Feed Tracking
Consumption difficult to track consistently

Milk & Quality
Production history and quality disconnected from collection

Sales & Payments
Buyer, sales, and payment info scattered across sources



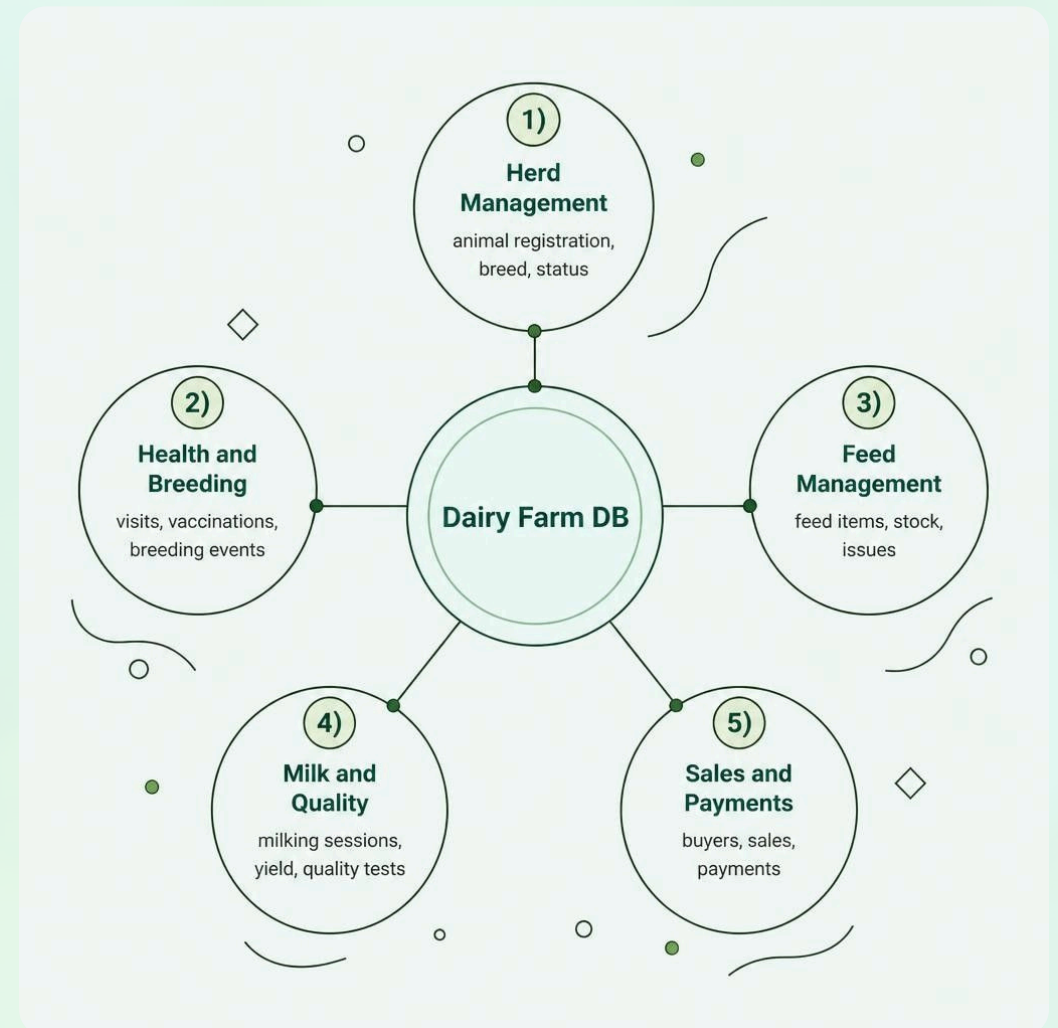
Disconnected data sources produce duplication, inconsistency, and reporting gaps.

One Database. One Source of Truth.

A centralized relational database unifies all farm operations—creating full traceability from animal → milk collection → quality → sale → payment.

Five integrated modules share a single schema, eliminating silos and enabling end-to-end operational reporting.

Traceability chain: **Animal → Milking → Quality → Sale → Payment**



From Herd Management to Revenue.

01 · Herd Management

- Animal registration
- Breed information
- Animal status tracking

02 · Health & Breeding

- Health visits & diagnosis
- Vaccination records
- Breeding events

03 · Feed Management

- Feed items & stock
- Feed issue records
- Stock tracking

04 · Milk Collection

- Milking sessions
- Milk yield recording
- Quality testing

05 · Commercial Management

- Buyer management
- Milk sales records
- Payment tracking

OBJECTIVES

Designed for Traceability, Accuracy & Control.



Centralize Farm Records

Single relational database for all operational data



Complete Animal History

Full lifecycle records from registration to status



Health & Vaccination Tracking

Structured visit, diagnosis, and vaccination records



Breeding Activity Monitoring

Track breeding events and reproductive history



Feed Usage & Stock

Monitor feed inventory and animal consumption



Milk Production & Quality

Accurate yield recording and quality test data



Sales & Payment Management

Manage buyers, sales, and payment transactions



Meaningful Reporting

Enable operational and commercial analysis

Who Interacts With the System?

Five distinct roles interact with the system, each with scoped access aligned to their operational responsibilities.

- **Farm Manager** — Overall monitoring, reports, farm records
- **Veterinarian** — Health visits, diagnosis, vaccinations
- **Farm Worker** — Animal records, feed issues, milking records
- **Milk Collection Staff** — Milk yield and quality records
- **Sales / Admin** — Buyers, sales, and payments

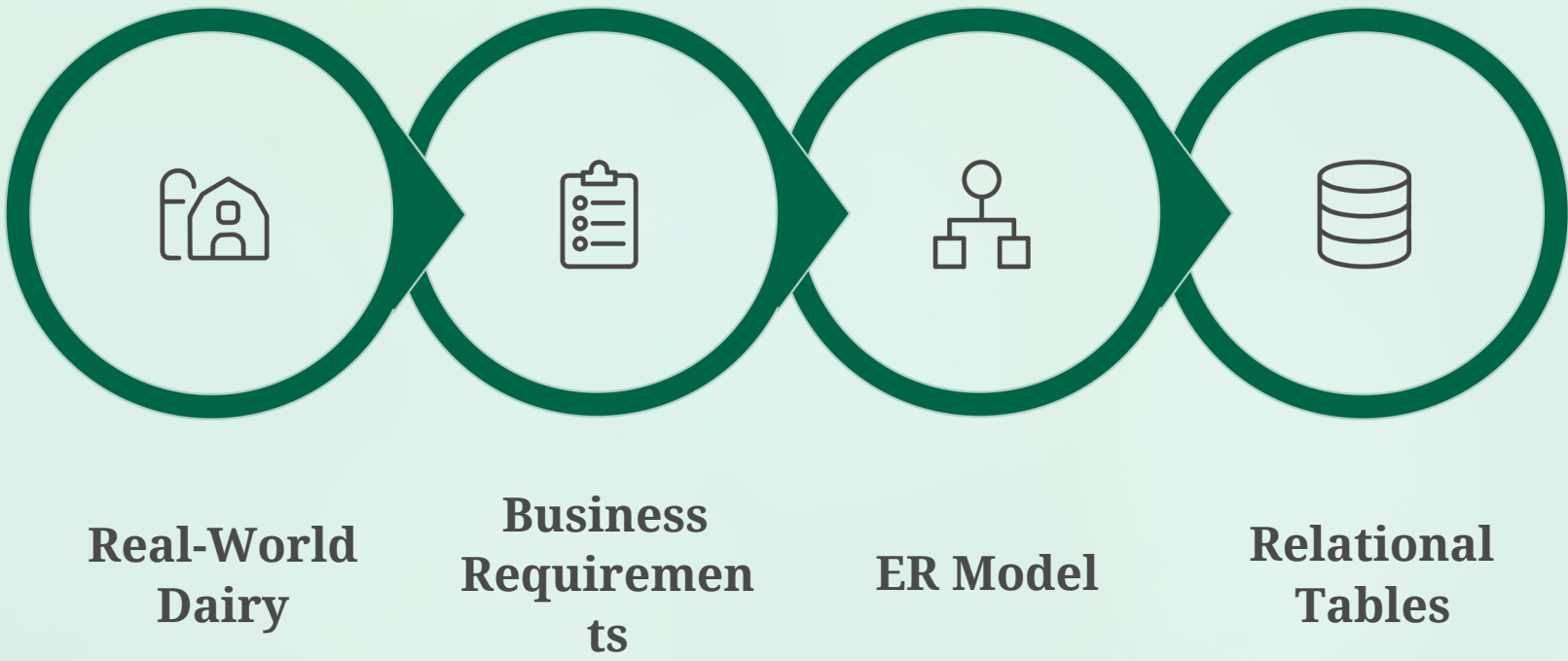


What Must the System Do?

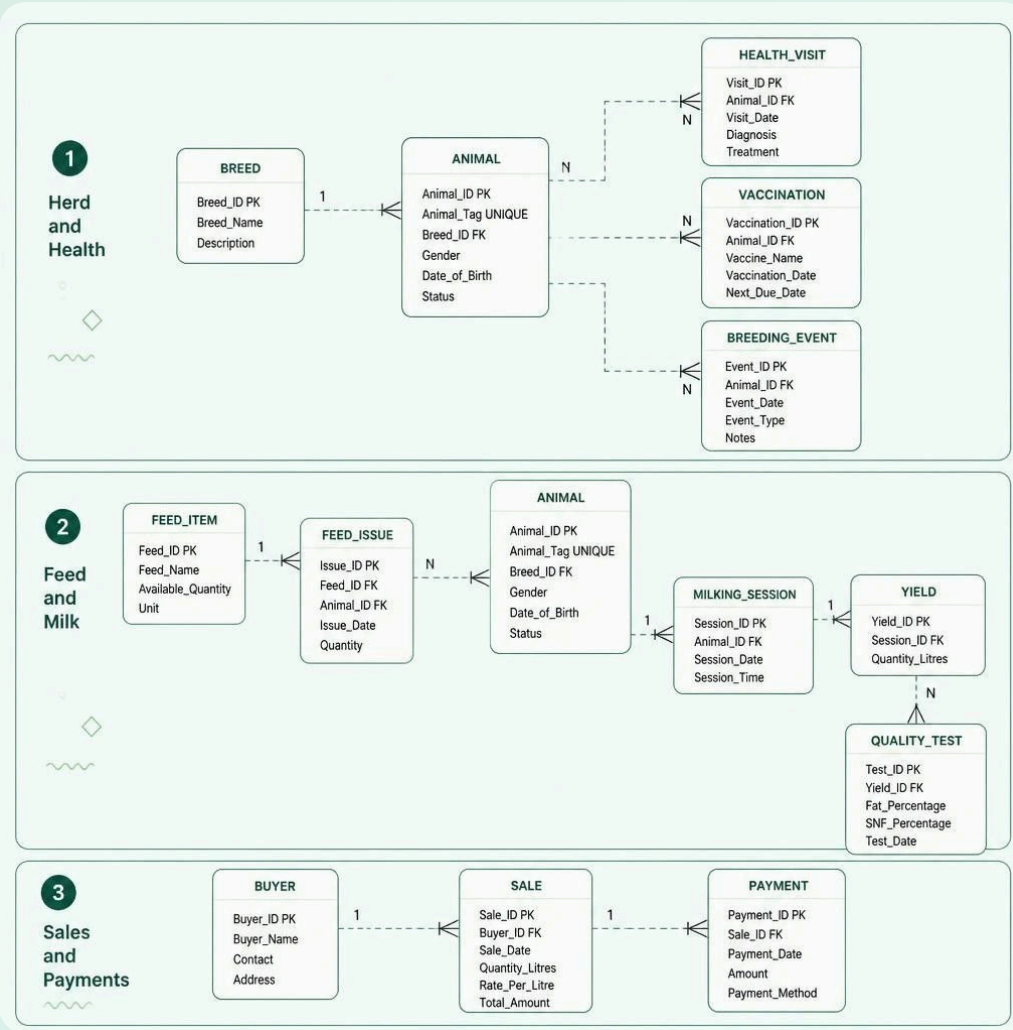
	Animal • Register & update animal • Search animal • Track status
	Health • Record health visits • Record vaccinations • Record treatments
	Breeding • Recordbreeding events • Track breeding history
	Feed • Maintain feed items • Record feed issues • Track available stock
	Milk • Record milking sessions • Record yields • Record quality tests
	Sales • Manage buyers • Record milk sales • Record payments

Conceptual ER Model

Threedataflows drive the system: **Herd History**(Breed → Animal → Health/Vaccination/Breeding), **Milk Traceability** (Animal → Milking Session → Yield → Quality Test), and **Commercial Flow** (Milk → Sale → Buyer → Payment). Feed Item connects to Feed Issue and Animal. These relationships create end-to-end traceability across all farm operations.



The ER model captures real-world entities, attributes, and relationships before implementation as relational tables.



Legend

- **PK** = Primary Key
- **FK** = Foreign Key
- **UNIQUE** = Unique Constraint

ER Model → Relational Tables

Each entity becomes a relational table. Relationships are enforced through foreign keys.

BREED

- Breed_ID **PK**
- Breed_Name
- Description

ANIMAL

- Animal_ID **PK**
- Animal_Tag UNIQUE
- Breed_ID **FK**
- Gender · Date_of_Birth · Status

HEALTH_VISIT

- Visit_ID **PK**
- Animal_ID **FK**
- Visit_Date · Diagnosis · Treatment

VACCINATION

- Vaccination_ID **PK**
- Animal_ID **FK**
- Vaccine_Name · Vaccination_Date · Next_Due_Date

BREEDING_EVENT

- Event_ID **PK**
- Animal_ID **FK**
- Event_Date · Event_Type · Notes

MILKING_SESSION

- Session_ID **PK**
- Animal_ID **FK**
- Session_Date · Session_Time

YIELD

- Yield_ID **PK**
- Session_ID **FK**
- Quantity_Litres

QUALITY_TEST

- Test_ID **PK**
- Yield_ID **FK**
- Fat_Percentage · SNF_Percentage · Test_Date

BUYER

- Buyer_ID **PK**
- Buyer_Name · Contact · Address

SALE

- Sale_ID **PK**
- Buyer_ID **FK**
- Sale_Date · Quantity_Litres · Rate_Per_Litre · Total_Amount

PAYMENT

- Payment_ID **PK**
- Sale_ID **FK**
- Payment_Date · Amount · Payment_Method

FEED_ITEM

- Feed_ID **PK**
- Feed_Name · Available_Quantity · Unit

FEED_ISSUE

- Issue_ID **PK**
- Feed_ID **FK** · Animal_ID **FK**
- Issue_Date · Quantity

Rules That Protect Data Integrity.

Uniqueness & Validity

- Animal tags must be **UNIQUE**
- Vaccination dates must be valid
- Required fields cannot be NULL

Non-Negative Values

- Milk yield cannot be negative
- Quality measurements cannot be negative

Referential Integrity

- Foreign keys must reference valid parent records
- Feed issue quantity $\leq$ available stock
- Milk sold $\leq$ milk collected